# 5CM510 – Network System development
# Lecture 18: 5G towards 6G Networks

UNIVERSITY OF DERBY

# Lecture Outline

- Fifth Generation (5G) of Wireless Networks
- 5G Radio and Technological components
- Massive MIMO
- Towards the Sixth Generation (6G) of Wireless Networks

# 5G and 6G: Evolution

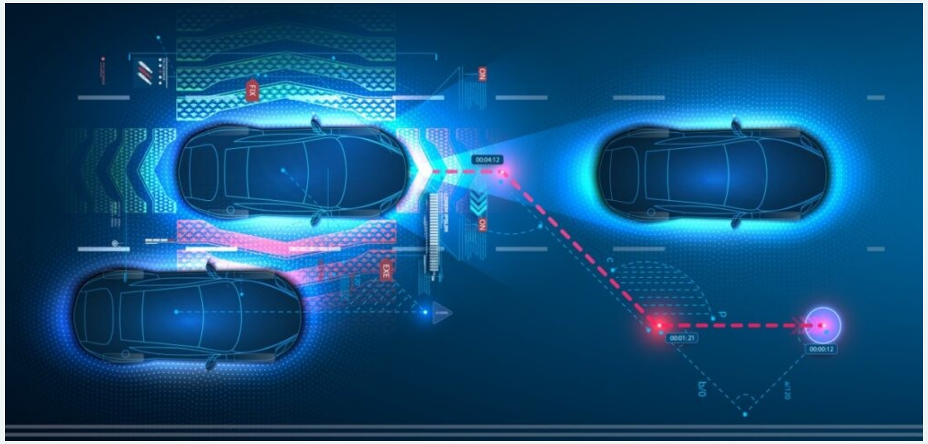
- Starting from the origin of mobile networks
- there was analogue circuit-switched 1G,
- then digital circuit-switched 2G,
- Digital packet-switched 3G, except for operator voice signals.
- With 4G, we see the move to 100% IP technology, even for voice signals, and 4G also handles multimedia applications.
- 5G offers increased data rates, critical missions and the connection of billions of devices.
- 6G is expected to be built on the strong foundation of 5G, in particular its ultra-high speed and reliability with ultra-low latency. 6G envisions four-tier network architectures that will extend the 5G space-air-ground networks by integrating underwater networks and incorporating key enabling technologies such as millimetre-wave and Terahertz communications as well as brand-new wireless communication technologies, most notably reconfigurable intelligent surfaces (RIS)

# 5G and 6G: Evolution

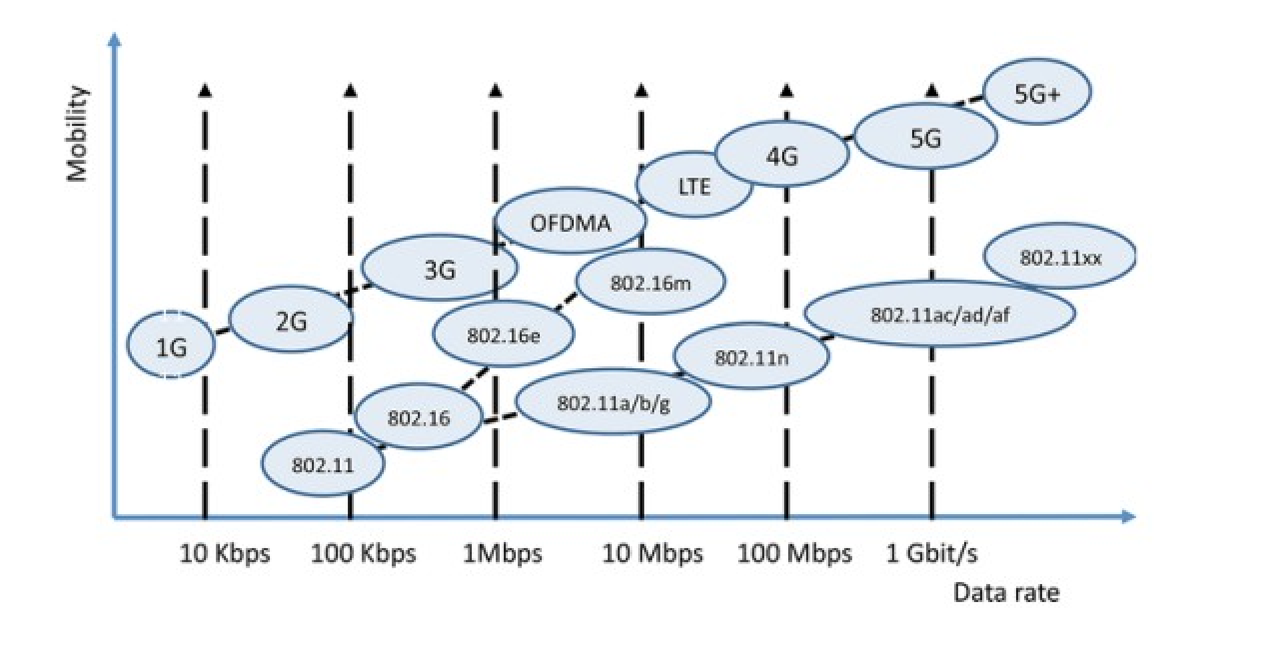
Question: What are the limitations of 4G that 5G and 6G aim to solve?

Industry Example: Autonomous Vehicles

5G enables real-time communication between cars, reducing latency in vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication.



# Generations of Wireless Networks



# 5G: What is  Changing?

- Which generation first introduced packet-switched networks?
- What is the primary benefit of 5G over 4G?
- Industry Example: Smart Cities
- How 5G supports real-time monitoring, smart grids, and automated traffic management.




Sensitivity: Internal

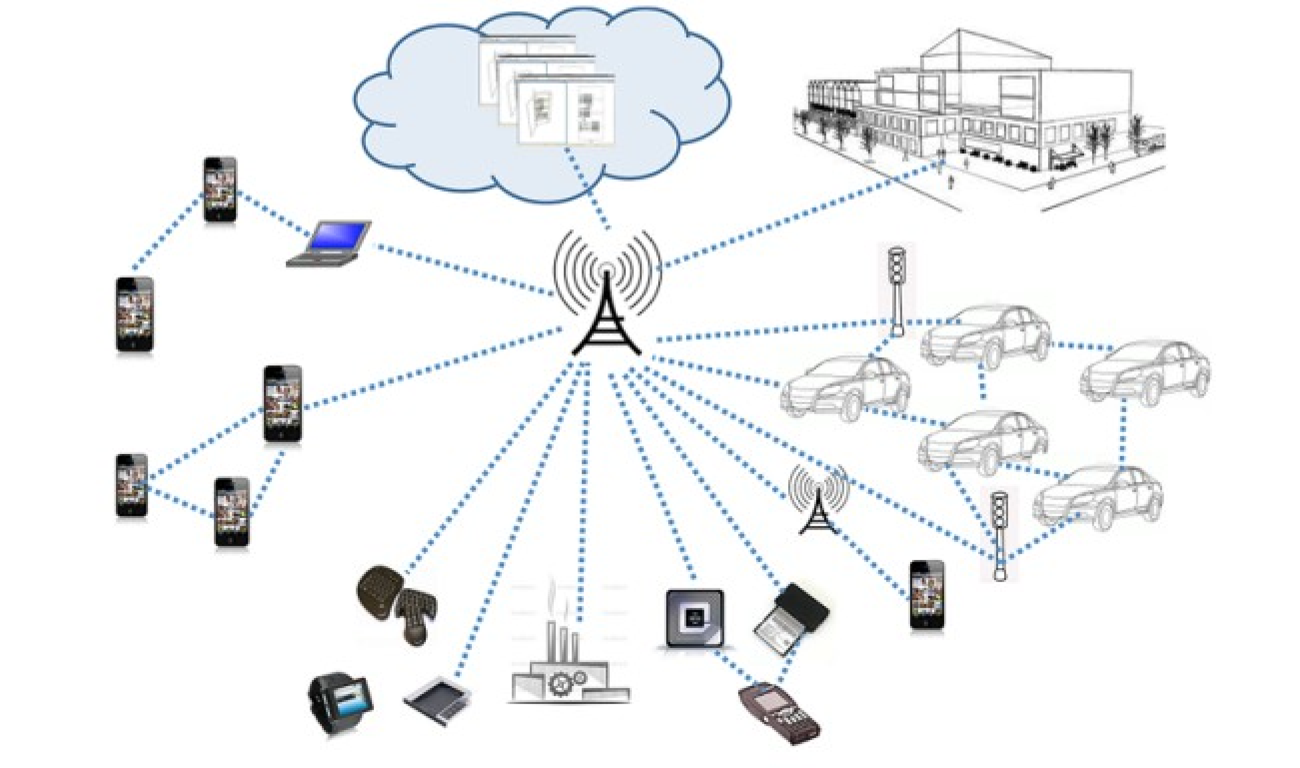
# 5G: Background

- 5G is 4G's successor. It is likely to be 2020 before it is standardised, and a few years after that before it begins being commercialised.
- The arrival of 5G was a two-phase process. The first phase, release 15, is the radio part, also called NR (New Radio) and was finalized in mid-2018.
- The second phase, release 16, defines the core network. Its standardization will end in June 2020. Together, these two standards form 5G. There are many developments in defining 5G from the radio aspect, and a development for the core network. Let us start with the developments for the radio.

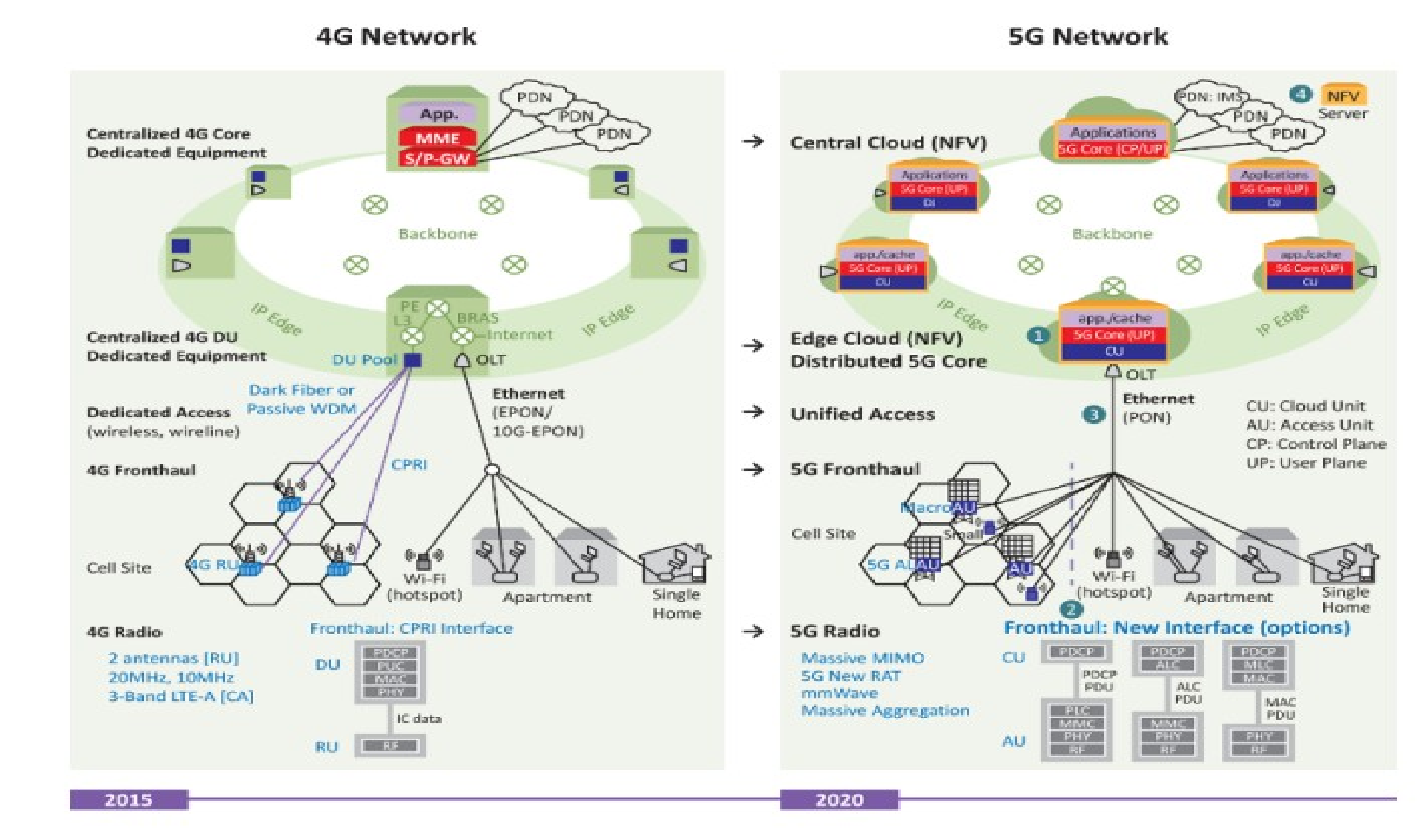
# 5G Access Scenarios

- multi-hop communications: the terminal is too far away from the gNodeB (Base station), and the signals pass through intermediary machines. This solution helps to limit the number of antennas while covering a large geographical area. Mesh networks or ad hoc networks provide this type of service, using algorithms to route the signals between the different terminals.
- D2D (Device to Device) – i.e. direct communication between two terminals without going through an antenna. This is a very eco-friendly solution, as it saves energy, minimizing the data rate of the system and thus decreasing electromagnetic pollution.
- M2M (Machine-to-Machine) or M2H (Machine-to-Human) communication. The connection of things is also represented by medical devices, household appliances or other items. energy expenditure depends on the square of the distance, it is very much to our advantage to use devices with short ranges.
- The problem of high density – e.g. the coverage of a stadium holding 100,000 people, with each spectator being able to enjoy a data rate of approximately 10 Mbps. The total data rate of 1 Tbps, in this scenario, could easily be delivered by an optical fiber, but the difficulty lies in the distribution of that capacity wirelessly.

# 5G Access Scenarios



# From 4G to 5G



4G Network
5G Network
Centralized 4G Core
Dedicated Equipment
App.
MME
S/P-GW
PDN
Backbone
IP Edge
Centralized 4G DU
Dedicated Equipment
DU Pool
BRAS
Internet
OLT
Dedicated Access
(wireless, wireline)
Dark Fiber or
Passive WDM
Ethernet
(EPON/
10G-EPON)
4G Fronthaul
CPRI
Cell Site
4G RU
Wi-Fi
(hotspot)
Apartment
Single
Home
4G Radio
2 antennas [RU]
20MHz, 10MHz
3-Band LTE-A [CA]
Fronthaul: CPRI Interface
DU
RU
IC data
Central Cloud (NFV)
Applications
5G Core (CP/UP)
PDN: IMS
NFV
Server
app./cache
5G Core (UP)
CU
Edge Cloud (NFV)
Distributed 5G Core
Unified Access
Ethernet
(PON)
CU: Cloud Unit
AU: Access Unit
CP: Control Plane
UP: User Plane
5G Fronthaul
Macro AU
Small
5G AU
AU
5G Radio
Massive MIMO
5G New RAT
mmWave
Massive Aggregation
Fronthaul: New Interface (options)
PDCP
PDU
ALC
PDU
MAC
PDU
2015
2020

# 5G Radio

- The 5G network has a rather heterogeneous radio to take care of rather different services, such as the connection of objects requiring particularly low data rates and connections to smartphones, for instance, on at a very high-speed moving train.
- Radio requires a set of techniques and frequency bands going from very low frequencies to frequencies of 120 GHz.
- The use of millimetre waves above 20 GHz allows us to go towards very high data rates and unify the techniques used in users' access and backhaul.
- Radio access can also use open bands, that is without the need for a license, by considering the regulation linked to open bands.
- It is possible to have bands with an exclusive license, a non-exclusive license or no license at all.

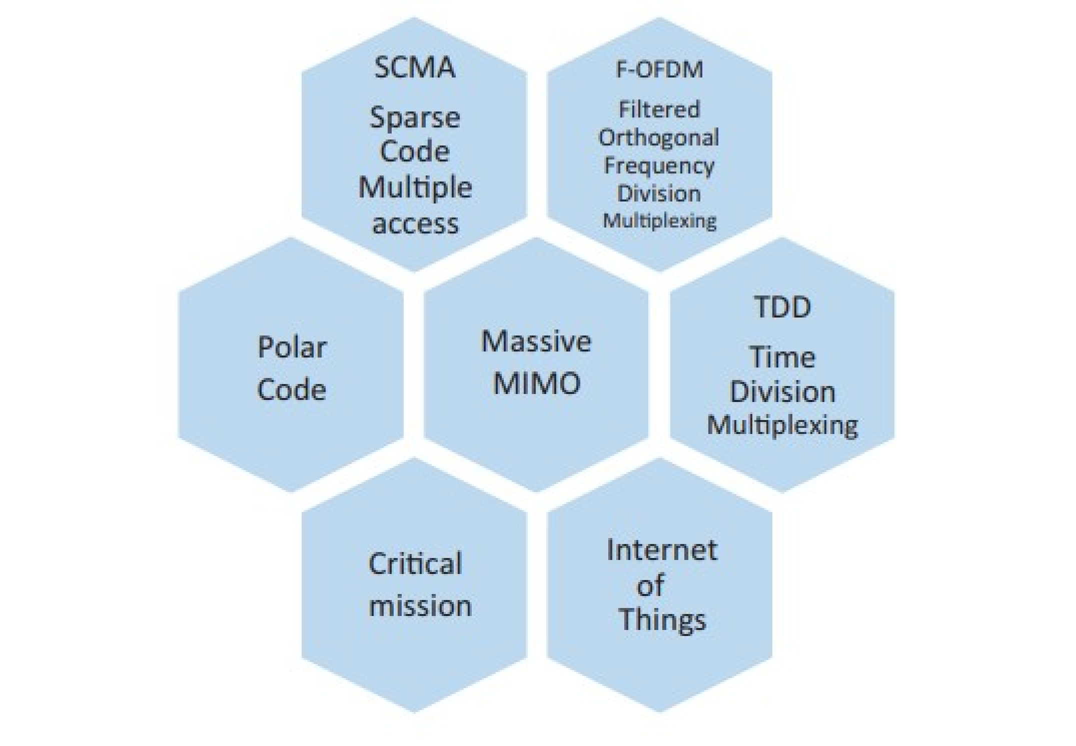
# 5G Use Cases – Real World Impact

Question: How does 5G impact healthcare, transport, and entertainment?

Industry Example: Healthcare – Remote Surgery

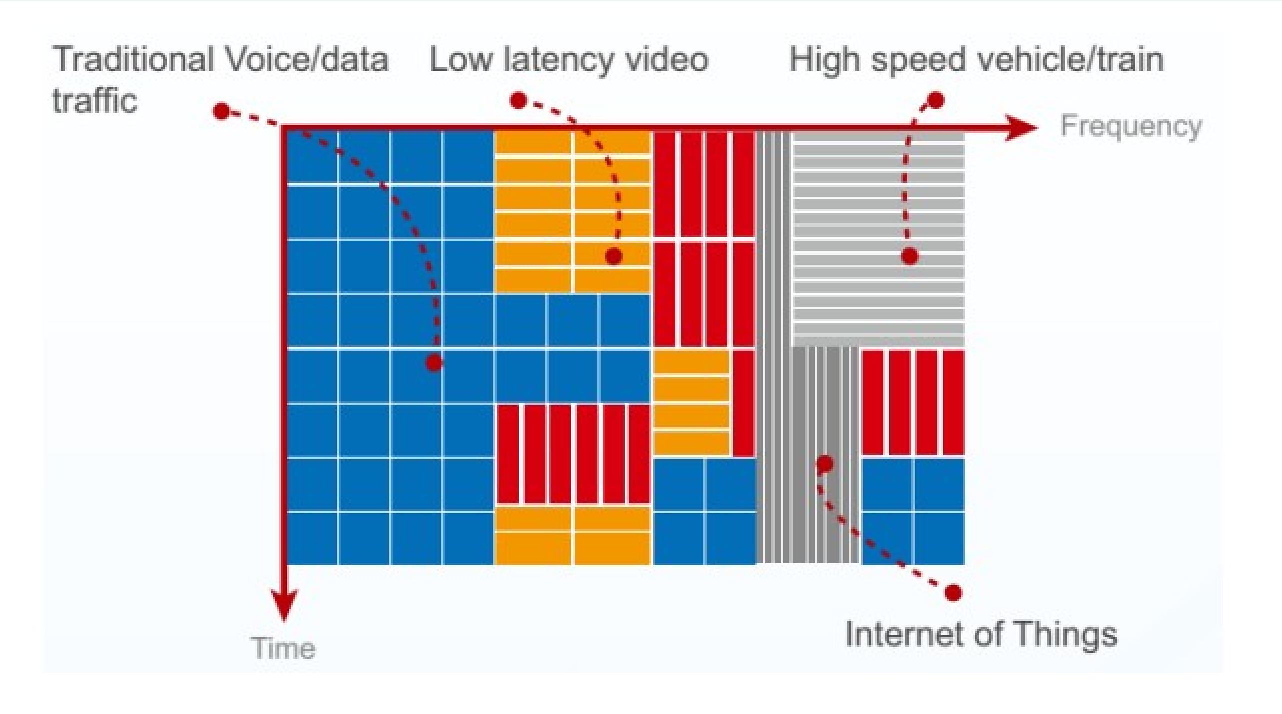
5G enables surgeons to perform operations remotely using ultra-low latency networks.

# Technological components of 5G



# Filtered OFDM

- The F-OFDM (Filtered-OFDM) is also a key element of 5G.
- It simultaneously exploits several subcarriers in each frequency band. Technology defines the shape of the waves.
- More exactly, it defines the different shapes of waves, access schemes and frame structures depending on application scenarios and needs to perform a service.
- The solution allows coexisting of different shapes of waves linked to OFDM parameters.



# SCMA (Sparse Code Multiple Access)

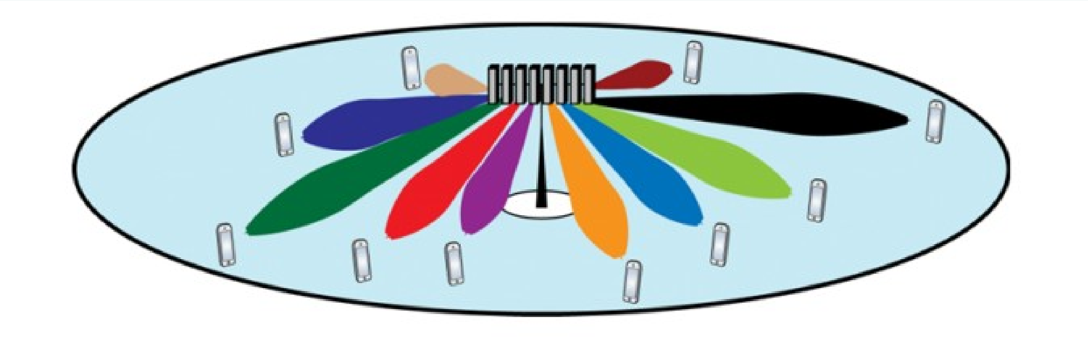
- SCMA (Sparse Code Multiple Access) is a new configuration of wave shape for the new radio interface which allows **more flexibility**.
- The shape of this wave, which is not orthogonal, facilitates a new multiple access scheme in which several code words, coming from different peripheral devices, are overlaid in code and power domains and carried on shared time-frequency resources.
- Multiplexing of several peripheral devices can become ineffective if the number of layers that are overlaid is superior to the length of the multiplexed code words. However, with SCMA, overload is tolerable with a significant detection complexity.
- In the SCMA, coded bits directly correspond with code words selected in a list of specific codes linked to the layer. The complexity of the detection highly depends on the number of words of the code list.

# Polar codes

- Polar codes are another major breakthrough for 5G.
- They form a class of **error correcting codes** discovered in 2009. They can reach the maximum capacity of the channel defined by Shannon with simple encoders and decoders when the size of the block is large enough.
- Polar codes have raised a lot of interest, and many research studies have been done, mainly on code building and decoding algorithms. A very positive aspect of this technique is its lack of complexity to encode and decode.
- This allows us to use encoding and decoding in real time on very high data rates.
- One of the most important decoding algorithms is the SC (Successive Cancellation) list decoding.
- Many performance evaluations show that polar codes concatenated with cyclic redundancy code (CRC) and an adaptive SC list decoder can overcome the turbo codes used in 4G. Thanks to polar codes, 5G clearly overtakes 4G performance.

# Massive MIMO

- The last great novelty of 5G is the use of **Massive MIMO**. This technique uses a very high number of antennas to link M antennas to K peripheral devices.
- In this case, M is very large compared to K.
- From a commercial point of view, massive MIMO is an attractive solution because its efficiency can be 100–1,000 times higher than that of 4G techniques.
- The idea is to perform several simultaneous communications using the same time-frequency resource. Spectral efficiency is much stronger thanks to this technique.

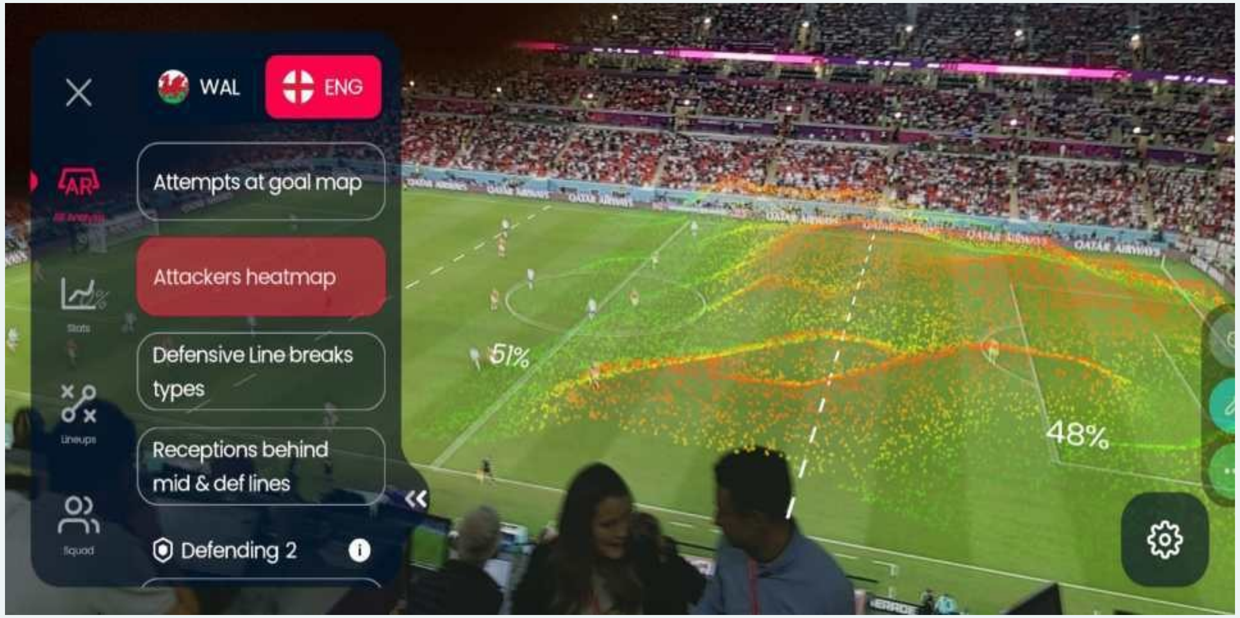


# Massive MIMO in Action

Question: Why is Massive MIMO important for high-density areas like stadiums and urban centres?

Industry Example: 5G in Sports Venues

Enhanced live streaming and real-time AR experiences in stadiums.






Sensitivity: Internal

# Time Division Multiplexing

- TDM is a core concept in telecommunications that allows multiple data signals to be transmitted simultaneously over a single communication channel.
- It operates by dividing the channel into distinct time slots.
- Each signal is assigned a specific time slot, which it uses to transmit its data.
- This method ensures that the signals transmitted do not interfere with each other, much like how scheduled times for trains prevent collisions on a single track.
- The ultra-fast speeds and low latency offered by 5G demand more efficient data handling methods, where advanced TDM techniques can play a crucial role.
- By optimising time slot allocation and dynamically adjusting to network conditions, TDM can significantly bolster the performance of next-generation networks.

# 6G Networks: Four-tier Networks

- 6G network architectures are anticipated to extend the 5G three-tier space-air-ground networks by integrating underwater networks, thus giving rise to four-tier space-air-ground-underwater networks with near-instant and unlimited super connectivity in the sky, at sea, and on land.
- **Space-network tier**: This network tier will support orbit or space Internet services in such applications such as space travel and provide wireless coverage via satellites.
- For long-distance intersatellite transmission in free space, laser communications represents a promising solution.
- The use of mmWave frequencies to establish high-capacity (inter)satellite communications may be another feasible solution to complement terrestrial 6G networks with computing stations placed on satellite platforms.
- **Air-network tier**: This network tier works in the low-frequency, microwave, and mmWave bands to provide more flexible and reliable connectivity for urgent events or in remote areas by densely employing flying base stations, e.g. unmanned aerial vehicles (UAVs).

# 6G Networks: Four-tier Networks

- **Terrestrial-network tier**: This network tier will still be the main solution for providing wireless coverage for most human activities. It will support low-frequency, microwave, mmWave, and THz bands in ultra dense heterogeneous networks, which require the deployment of ultra-high-capacity backhaul infrastructures.
- **Underwater-network tier**: This network tier will provide coverage and Internet services for broad-sea and deep-sea activities for military or commercial applications. Given that water exhibits different propagation characteristics, acoustic and laser communications can be used to achieve high-speed data transmission for bidirectional underwater communication

# 6G Networks: The Future of Connectivity

**Question: What new possibilities does 6G introduce beyond 5G?**

**Industry Example: Holographic Telepresence**

**6G enables real-time, high-resolution holographic video calls with ultra-low latency.**

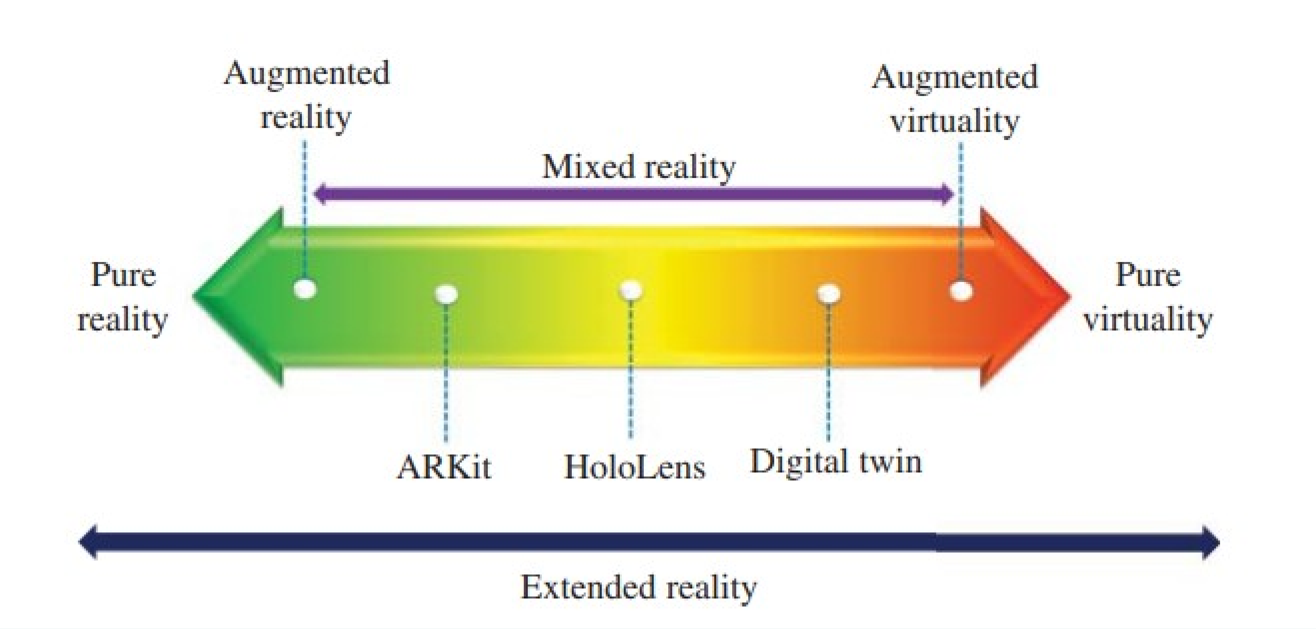
# 6G : Enabling Technologies

- **Millimetre-Wave and Terahertz Communications**: Higher frequencies from 100 GHz to 3 THz are promising bands for the next generation of wireless communication systems, offering the potential for revolutionary applications. Technically, the formal definition of the THz region is 300 GHz through 3 THz, though sometimes the terms sub-THz or sub-mmWave are used to define the 100–300 GHz spectrum. The short wavelengths at mmWave and THz will allow massive spatial multiplexing in hub and backhaul communications.
- The THz band from 100 GHz through 3 THz can enable secure communications due to the fact that small wavelengths allow for extremely high-gain antennas with extremely small physical dimensions.
- **Reconfigurable Intelligent Surfaces:** A brand-new wireless communication technology referred to as reconfigurable intelligent surfaces (RISs) – also known as large intelligent surfaces, smart reflect-arrays, intelligent reflecting surfaces, passive intelligent mirrors, artificial radio space, or programmable meta surface.
- RISs are often referred to as software-defined surfaces(SDSs) in analogy with the concept of software-defined radio (SDR).

# 6G : Enabling Technologies

- **From Network softwarization to Network intelligence:** 6G will take network softwarization to a new level, namely toward network intelligentization. Software-defined networking (SDN) and network function virtualization (NFV) have moved modern communications networks toward software-based virtual networks.
- They also enable network slicing, which can provide a powerful virtualization capability to allow multiple virtual networks to be created atop a shared physical infrastructure.
- As the network is becoming more complex and more heterogeneous, softwarization is not going to be sufficient for 6G. Existing technologies such as SDN, NFV, and network slicing will need to be further improved by enabling fast learning and adaptation via AI-based methods.
- Approaches like network slicing will become much more versatile and intelligent to support diverse capabilities and more advanced IoT functionalities, including sensing, data collection, analytics, and storage.

# Qualcomm: Extended Reality (XR)





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# 6G Vision : Mixed reality (MR)

- A good AR example is Shopify's ARKit, which allows smartphones to place 3D models of physical items and see how they would look in real life. To do so, AR overlays virtual objects/images/information on top of a real-world environment.
- An illustrative augmented virtuality (AV) example is an aircraft maintenance engineer, who can visualize a real-time model, often referred to as digital twin, of an engine that may be thousands of kilometres away.
- The term mixed reality (MR) includes AR, AV, and mixed configurations thereof, blending representations of virtual and real-world elements together in a single user interface.
- MR helps bridge the gap between real and virtual environments, whereby the difference between AR and AV reduces to where the user interaction takes place. If the interaction happens in the real world, it is considered AR.

# Summary

- 5G is currently being deployed, but industries should prepare for 6G.
- AI, Quantum Computing, and Edge Computing will play key roles in 6G development.
- Businesses and researchers must focus on network intelligence, security, and sustainable communication technologies.
- Features of 5G - Higher Data Rates: Multi-gigabit speeds for real-time applications, Low Latency: Essential for mission-critical applications like remote surgery, Massive Connectivity: Supports billions of devices, enabling smart cities and IoT, Network Slicing: Customizes network functions for different use cases.
- Future Applications of 6G - Holographic Communication: Real-time, high-definition virtual meetings, Smart Factories: Fully automated AI-driven manufacturing, Deep-Sea & Space Connectivity: Global seamless communication.


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